On Perfect Functional Representations

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***Abstract*—For given random variables (X, Y), functional representation lemma states that there exists a random variable Z and a deterministic function g such that $Y = g(X, Z)$, and X is independent of Z. In other words, Z captures all the information about Y not contained in X. We refer to Z as a 'perfect functional representation' if it satisfies $I(Y; Z) = H(Y|X)$. We show that the existence of perfect representation depends solely on the conditional distribution $p_{Y|X}$ and define the corresponding conditional distributions as perfectly representable. We provide a necessary and sufficient condition for $p_{Y|X}$ to be perfectly representable. Finally, we cast the problem of checking perfect representability as a linear program and discuss its computational tractability.**

I. Introduction

For given random variables (X, Y), functional representation lemma [1] states that there exists a random variable Z and a deterministic function g such that $Y = g(X, Z)$, and X is independent of Z. In other words, the triple (X, Y, Z) satisfies

$$I(X; Z) = 0 \quad \text{and} \quad H(Y|X, Z) = 0. \tag{1}$$

A random variable Z that satisfies (1) is not unique. The problem of finding a Z with minimum entropy has been studied extensively in [2]–[13] and references therein. The problem of finding Z maximizing the mutual information with Y has been a key technical ingredient in information theory problems such as channel synthesis [14]–[17], and many others [14], [15]. Finally, there are many interesting connections between functional representation and information-theoretic privacy [5], [6], [18]–[26].

Of particular interest is the problem of finding a Z that is as informative as possible about Y in terms of mutual information. That is, the goal is to maximize $I(Y; Z)$ under the constraints in (1). Observe that by the chain rule,

$$I(Y; Z) = H(Y|X) - I(X; Z|Y) - H(Y|X, Z) + I(X; Z), \tag{2}$$

and by (1) we have

$$I(Y; Z) = H(Y|X) - I(X; Z|Y) \leq H(Y|X). \tag{3}$$

Hence, the maximum mutual information is trivially upper-bounded by $H(Y|X)$, with equality whenever $I(X; Z|Y) = 0$. This happens if and only if $X - Y - Z$, i.e., the triple (X, Y, Z) forms a Markov chain.

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It is shown in [14], [16] that it is always possible to achieve the upper bound in (3) within a logarithmic factor. That is, there exists a Z such that

$$H(Y|X) - \log(I(X; Y) + 1) - 4 \leq I(Y; Z). \tag{4}$$

From this, we can easily see that the trivial upper bound in (3) is achieved asymptotically for product distributions. That is, applying the construction in (4) to (X^n, Y^n) gives the term $I(X^n; Z_n|Y^n)$ which is on the order of $\log n$ (and this goes to zero if we normalize by n). This problem is further studied in the context of channel synthesis in [17] where it is shown that the redundancy term $I(X^n; Z_n|Y^n)$ is $\frac{1}{2}\log n$ for (X^n, Y^n) product pairs with a so-called non-singular conditional distribution $p_{Y|X}$. Moreover, for product pairs with a singular conditional distribution, $p_{Y|X}$, the redundancy term is shown to be sub-logarithmic.

The goal of this paper is to characterize random variables for which there exists a functional representation Z such that the trivial upper bound in (3) is achieved exactly. We call such pairs (X, Y) perfectly representable and Z a perfect functional representation of the pair (X, Y). This problem is surprisingly ubiquitous. As it is mentioned in [5], there exist a perfect functional representation in some special cases: for example, if X is a function of Y or Y is a function of X. In [14] it is shown that the term $I(X; Z|Y)$ is minimized by the Poisson functional representation whenever the support of Y is two. This problem was further studied in [18] where it is shown that if Wyner's common information for (X, Y) is the same as the mutual information, then the pair is perfectly representable. Moreover, it could be observed from [27], [28] that perfect representability is equivalent to $H_K(G_{YX}, Y) = H(Y|X)$ where H_K denotes Körner graph entropy and the graph G_{YX} is defined as in [29]. Likewise, perfect representability is equivalent to perfect privacy in the privacy funnel, i.e., $G_{\mathrm{PF}}(H(Y|X), X \to Y) = 0$ (see [30]), or equivalently $g_0(X; Y) = H(Y|X)$ (see [31]). Finally, this problem could be viewed as a special case of the channel synthesis problem in [17]. That is, we characterize exactly when the redundancy term in channel synthesis is zero, and not just vanishing asymptotically.

The remainder of this paper is structured as follows. In II, we define and give some initial results on perfectly representable conditional distributions. In III, we provide our main results which is a necessary and sufficient condition for perfect representability. In IV, we present an algorithm for checking if the conditional distribution is perfectly representable. Some of the proofs are omitted due to space constraints.

II. Preliminaries

Let $\mathcal{X}$ and $\mathcal{Y}$ denote the support of X and Y respectively. We assume, without loss of generality, that $\mathcal{X} = \{1, 2, ..., k\}$ and $\mathcal{Y} = \{1, 2, ..., m\}$ for some $k, m \in \mathbb{Z}^+$.

A. Basic Definitions

Given a functional representation Z of (X, Y) such that $Y = f(X, Z)$, the vector

$$W = \left[f(1, Z)\ f(2, Z)\ \ldots\ f(k, Z)\right]^T \tag{5}$$

has the following properties:

1) W is independent of X.
2) With $g(X, W) := W_X$ (i.e. X-th entry of the random vector W), we have $Y = g(X, W)$, consequently, W is also a functional representation of (X, Y).
3) $W_i \sim p_{Y|X=i}$ for each $i \in \{1, 2, \ldots, k\}$.
4) $I(Y; W) = I(Y; Z)$.

First three properties are trivial, and the last one is shown in Lemma 1 below. This observation motivates the following definition which is also a result of the Lemma 3.22 from [32].

Definition 1. *For a given $(X, Y) \sim p_{XY}$, a canonical functional representation of (X, Y) is a random vector $[W_1, W_2, ..., W_k]^T$ independent of X such that $W_i \sim p_{Y|X=i}$ for all $i \in \{1, 2, ..., k\}$.*

Remark 1. *If W is a canonical functional representation for (X, Y), it is also a canonical functional representation for all (X', Y') with conditional distribution $p_{Y|X}$ regardless of the distribution of X'.*

Lemma 1. *Given an (X, Y, Z) triple that satisfies (1), there is a canonical functional representation W for the pair (X, Y) that satisfies $I(Y; W) = I(Y; Z)$.*

Proof. Choose the random variable W as

$$W = \left[f(1, Z)\ f(2, Z)\ \ldots\ f(k, Z)\right]^T \tag{6}$$

where f is such that $Y = f(X, Z)$. Then, W is a function of Z. By applying the equality in (3) to both Z and W, we get

$$I(Y; Z) - I(Y; W) = I(X; W|Y) - I(X; Z|Y). \tag{7}$$

Since W is a function of Z, the left hand side of (7) is non-negative and the right hand side of (7) is non-positive. Consequently, both sides are equal to zero. □

Note that Lemma 1 implies that if there exists a functional representation Z of $(X, Y) \sim p_{XY}$, then there exists a canonical functional representation W that is as informative as Z about Y. Thus, it is sufficient to focus on canonical representations.

Definition 2. *A conditional distribution $p_{Y|X}$ is said to be perfectly representable if there exists $p_{W|Y}$ such that W is a canonical functional representation for (X, Y).*

If $p_{XYW} = p_{W|Y} p_{Y|X} p_X$, then $X - Y - W$ is a Markov chain. If W is a canonical functional representation, then $I(Y; W)$ satisfies (3) with equality. From Remark 1, we see that (X, Y) being perfectly representable or not depends on $p_{Y|X}$ only. Hence, our main objective is reduced to investigating the family of conditional distributions $p_{Y|X}$ that are perfectly representable.

B. Singular Channels

Definition 3. *A discrete channel $p_{Y|X}$ is called singular if, for all $y \in \mathcal{Y}$ and all $x_1, x_2 \in \mathcal{X}$ with $p_{Y|X}(y|x_1)p_{Y|X}(y|x_2) > 0$, we have $p_{Y|X}(y|x_1) = p_{Y|X}(y|x_2)$. Moreover, we call a conditional probability matrix singular if it corresponds to a singular channel.*

Just like in [17], the concept of singular channels is important for our problem. As it turns out, singularity is a necessary condition for perfect representability.

Lemma 2. *If a conditional distribution $p_{Y|X}$ is perfectly representable, it is singular. Moreover, if $p_{Y|X}$ is singular and $|\mathcal{X}| = 2$ or $|\mathcal{Y}| = 2$, then $p_{Y|X}$ is perfectly representable.*

According to Lemma 2, singularity is a necessary condition for a channel to be perfectly representable. It is also a sufficient condition when $|\mathcal{X}| = 2$ or $|\mathcal{Y}| = 2$. However, it is not a sufficient condition in general.

Example 1. *For $|\mathcal{X}| = |\mathcal{Y}| = 3$, the channel*

$$p_{Y|X} = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 1/2 & 1/2 \end{bmatrix} \tag{8}$$

is singular, but not perfectly representable.

C. Total variation distance

The total variation distance of two distributions over the same alphabet is defined as

$$\mathrm{TV}(p_{X_1}, p_{X_2}) = \frac{1}{2}\sum_{x \in \mathcal{X}} |p_{X_1}(x) - p_{X_2}(x)|. \tag{9}$$

When $|\mathcal{X}| = 2$, we can find the functional representation Z, for any p_{XY}, that achieves the maximum $I(Y; Z)$ and calculate the value of $I(Y; Z)$ in terms of the total variation distance between the conditional distributions.

Lemma 3. *Given $(X, Y) \sim p_{XY}$ with $|\mathcal{X}| = 2$, define*

$$r = TV(p_{Y|X=1} || p_{Y|X=2}). \tag{10}$$

The maximum value of $I(Y; Z)$ among the all functional representations Z equals to $H(Y) - rH(X)$. Moreover, $rH(X)$ equals to $I(X; Y)$ if and only if $p_{Y|X}$ is singular.

III. Main Results

In this section, we give the necessary and sufficient condition for a channel to be perfectly representable. Our approach is to define four basic operations on a singular conditional distribution matrix. Three of these operations preserve the property of being perfectly representable and three of them preserve the property of not being perfectly representable. We show that the set of all perfectly representable channels are exactly the channels which could be obtained via those three operations that preserve the property of being perfectly representable from the trivial channel that maps every input to the same output.

A. Four Basic Operations

We can think of the channel $p_{Y|X}$ as a conditional probability matrix M of size $k \times m$ where $k = |\mathcal{X}|$ and $m = |\mathcal{Y}|$. Note that the i'th column of M is $p_{Y|X}(i|\cdot)$. By Lemma 2, all perfectly representable matrices must be singular. Since singularity is easy to check, we focus on singular matrices from now on.

Definition 4. *For a singular conditional probability matrix M, the unique non-zero value of the non-zero entries in a column is called the "value" of that column. Moreover, a column divided by its value β is called the "pattern" of that column. So, the pattern of column i is the vector $M[:, i]/\beta$.*

In Example 1, the first column of the singular matrix has value $\beta = 1/2$ and pattern $[1, 1, 0]^T$.

We define four operations on singular matrices. Note that it is straightforward to see that the result of these operations will be singular matrices as well. Let $\mathcal{S}$ denote the set of all singular matrices and $\mathcal{S}_{k,m}$ denote the set of singular matrices of size $k \times m$.

1) Pattern preserving split (PPS): Let $\alpha \in (0, 1)$ and $l \in \{1, 2, ..., m\}$. The pattern preserving split operation $A_{l,\alpha} : \mathcal{S}_{k,m} \to \mathcal{S}_{k,m+1}$ is the operation that takes a singular matrix M and splits the l'th column into two columns (that will correspond to l'th and $(l+1)$'th columns in the resulting matrix $A_{l,\alpha}(M)$) that are α and $(1 - \alpha)$ times the original column.

Example 2. *Let*

$$M = \begin{bmatrix} 1-\beta & \beta & 0 & 0 & 0 \\ 0 & \beta & 1-\beta & 0 & 0 \\ 1-\beta & 0 & 0 & \beta/2 & \beta/2 \end{bmatrix}, \tag{11}$$

then,

$$A_{2,1/2}(M) = \begin{bmatrix} 1-\beta & \beta/2 & \beta/2 & 0 & 0 & 0 \\ 0 & \beta/2 & \beta/2 & 1-\beta & 0 & 0 \\ 1-\beta & 0 & 0 & 0 & \beta/2 & \beta/2 \end{bmatrix}. \tag{12}$$

The pattern preserving split operation is applied to the second column of M with the splitting ratio of 1/2.

2) Pattern preserving merge (PPM): The pattern preserving merge operation $B_{l_1,l_2} : \mathcal{S}_{k,m+1} \to \mathcal{S}_{k,m}$ for some distinct $l_1, l_2 \in \{1, 2, ..., m\}$ is the operation that takes a singular matrix M and merges l_1'th and l_2'th columns (assuming that they have the same pattern). The new column will correspond to $\min\{l_1, l_2\}$'th column in the resulting matrix $B_{l_1,l_2}(M)$ that equals to the sum of these two columns (i.e. $B_{l_1,l_2}(M)[i, l] = M[i, l_1] + M[i, l_2]$ for all $i \in \{1, 2, ..., k\}$). Then, it shifts the columns on the right side of $\max\{l_1, l_2\}$'th column to the left and leaves the remaining columns unchanged to get the resulting matrix.

Example 3. *For the same M as in Example 2,*

$$B_{4,5}(M) = \begin{bmatrix} 1-\beta & \beta & 0 & 0 \\ 0 & \beta & 1-\beta & 0 \\ 1-\beta & 0 & 0 & \beta \end{bmatrix}. \tag{13}$$

The pattern preserving merge operation is applied to the fourth and fifth columns of M.

3) Value preserving split (VPS): Let $l \in [1 : m]$ and $\mathbf{b} \in \{0, 1\}^k$. The value preserving split operation $C_{l,\mathbf{b}} : \mathcal{S}_{k,m} \to \mathcal{S}_{k,m+1}$ is the operation that takes a singular matrix and splits the l'th column (whose value is $\beta \in (0, 1)$ and whose pattern is $\mathbf{p} \in \{0, 1\}^k$) into 2 columns (that will correspond to l'th and $(l+1)$'th columns in the resulting matrix $C_{l,\mathbf{b}}(M)$). Both columns of the new matrix will have value β and patterns $\mathbf{p}_1 \in \{0, 1\}^k$ and $\mathbf{p}_2 \in \{0, 1\}^k$ respectively where $\mathbf{p}_1 = \mathbf{p} \cdot \mathbf{b}$ and $\mathbf{p}_2 = \mathbf{p} \cdot (\mathbf{1}_k - \mathbf{b})$. Note that $\cdot$ represents elementwise multiplication and $\mathbf{1}_k$ represents the vector of size k all whose entries are 1.

Example 4. *For the same M as in Example 2,*

$$C_{1,[1,1,0]^T}(M) = \begin{bmatrix} 1-\beta & 0 & \beta & 0 & 0 & 0 \\ 0 & 0 & \beta & 1-\beta & 0 & 0 \\ 0 & 1-\beta & 0 & 0 & \beta/2 & \beta/2 \end{bmatrix}. \tag{14}$$

The value preserving split operation is applied to the first column of M with the splitting vector of $[1, 1, 0]^T$.

4) Value preserving merge (VPM): Let $l_1, l_2 \in \{1, 2, ..., m\}$ be such that the l_1'th and l_2'th columns of M have the same value $\beta \in (0, 1]$ and patterns $\mathbf{p}_1 \in \{0, 1\}^k$ and $\mathbf{p}_2 \in \{0, 1\}^k$ that satisfy $\mathbf{p}_1 + \mathbf{p}_2 \in \{0, 1\}^k$.

The value preserving merge (VPM) operation $D_{l_1,l_2} : \mathcal{S}_{k,m+1} \to \mathcal{S}_{k,m}$ is the operation that takes a singular matrix M and merges l_1'th and l_2'th columns into a single column with the pattern $\mathbf{p}_1 + \mathbf{p}_2$ and the value β (that will correspond to $\min\{l_1, l_2\}$'th column in the resulting matrix $D_{l_1,l_2}(M)$).

Example 5. *For the same M as in Example 2,*

$$D_{1,3}(M) = \begin{bmatrix} 1-\beta & \beta & 0 & 0 \\ 1-\beta & \beta & 0 & 0 \\ 1-\beta & 0 & \beta/2 & \beta/2 \end{bmatrix}. \tag{15}$$

The value preserving merge operation is applied to the first and third columns of M.

We explain how the order of these operations can be manipulated in the following lemma.

Lemma 4. *For any sequence of PPS, VPS, PPM operations, there exists an equivalent cascade that consists of a sequence of PPS operations, followed by a sequence of VPS operations, and finally a sequence of PPM operations.*

B. Perfect Representability

We are now ready to find the family of channels that are perfectly representable. We start with a channel $p_{Y|X}$ that maps every $x \in \mathcal{X}$ to a unique $a \in \mathcal{Y}$. This channel has a conditional probability matrix which is the all ones vector of length k and is clearly perfectly representable (choose $Z = a$). By Lemma 5, all matrices that we obtained from this matrix with the repeated use of pattern preserving split, pattern preserving merge, and value preserving split operations are perfectly representable. Finally, we will prove that these conditional probability matrices are the only perfectly representable conditional probability matrices.

Theorem 1. *A singular $k \times m$ conditional probability matrix M is perfectly representable if and only if it is possible to obtain the vector $\mathbf{1}_k$ from M via a repeated use of PPM, PPS, and VPM operations defined in section III-A.*

Recall that $\mathbf{1}_k$ is the all ones vector of size k and that $\mathbf{1}_k$ is perfectly representable. The "if" part of the proof is a direct result of the following Lemmas.

Lemma 5. *The result of applying PPM, PPS, and VPS operations described in section III-A to a perfectly representable channel will also be a perfectly representable channel.*

Since PPM and PPS operations are inverses of each other, the result of applying PPM and PPS operations to a channel that is not perfectly representable will also be a channel that is not perfectly representable. Moreover, since VPM and VPS operations are inverses of each other, the result of applying VPM operation to a channel that is not perfectly representable will also be a channel that is not perfectly representable. Thus, Lemma 6 is a direct result of Lemma 5.

Lemma 6. *The result of applying PPM, PPS, and VPM operations described in section III-A to a channel that is not perfectly representable will also be a channel that is not perfectly representable.*

See the Appendix for the "only if" part of the proof of Theorem 1.

Remark 2. *By the equivalences noted in the introduction, Theorem 1 also characterizes exactly when*

$$\begin{aligned} H_K(G_{YX}, Y) &= H(Y|X), && \textit{(see [27]–[29])},\\ G_{\mathrm{PF}}(H(Y|X), X \to Y) &= 0, && \textit{(see [30])},\\ g_0(X;Y) &= H(Y|X), && \textit{(see [31])}. \end{aligned}$$

Equivalently, it characterizes exactly when the redundancy term in channel synthesis is zero (see [17]).

IV. Algorithmic Results

In this section, we will focus on the problem of checking if a conditional distribution $p_{Y|X}$ is perfectly representable .

Note that checking whether a matrix is a singular conditional probability matrix or not can be done in linear time. Thus, we can assume that we are working with singular conditional probability matrices. A matrix M is obtainable from $\mathbf{1}_k$ by PPS, PPM, VPS operations if and only if the matrix M' that is obtained from M by merging all columns with the same pattern is obtainable from $\mathbf{1}_k$ by PPS, PPM, VPS operations. Therefore, we can assume that each column of M has a distinct pattern. That is, it has $m \le 2^k - 1$ columns with distinct patterns $\mathbf{p}_1, \mathbf{p}_2, ..., \mathbf{p}_m$ with values $v_1, v_2, ..., v_m$ in $[0,1]$. Thus, our singular conditional probability matrix could be solely defined by the vector of values $\mathbf{v}$ and patterns $\mathbf{p}_1, \mathbf{p}_2, ..., \mathbf{p}_m$.

A. Primal LP

Consider the all possible sets of binary vectors of length k (other than all zeros vector) that add up to $\mathbf{1}_k$. Note that the addition here is in real numbers arithmetic, not modulo 2 arithmetics. Let's call these sets as $S_1, S_2, ..., S_R$ where R equals to the k-th Bell number which is the number of set partitions of a k-element set [33].

Example 6. *For $k = 3$, R will equal to 5 and our sets will be $S_1 = \{[1,1,1]^T\}, S_2 = \{[1,1,0]^T, [0,0,1]^T\}, ..., S_5 = \{[1,0,0]^T, [0,1,0]^T, [0,0,1]^T\}$.*

The question is whether we can assign some weights $x_1, x_2, ..., x_R$ in $[0,1]$ to these sets such that

$$\sum_{j:\mathbf{p_i} \in S_j} x_j = v_i, \ \forall i \in \{1,2,...,m\} \tag{16}$$

Assigning such weights $x_1, x_2, ..., x_R$ to the sets $S_1, S_2, ..., S_R$ is equivalent to:

1) First, from $\mathbf{1}_k$, obtain the matrix with columns whose values are $x_1, x_2, ..., x_R$ and patterns are all $\mathbf{1}_k$ by PPS operation.
2) Then, split the i-th column with respect to S_i by VPS operation for all $i \in \{1,2,...,R\}$.
3) Finally, merge all columns with the same pattern by PPM.

Then, by Lemma 4, doing any cascade of PPS, VPS, and PPM operations is equivalent to assigning weights $x_1, x_2, ..., x_R$ to the sets $S_1, S_2, ..., S_R$.

Now, let's define the matrix $H \in \{0,1\}^{m \times R}$ such that $H_{ij} = \mathbb{1}\{\mathbf{p}_i \in S_j\}$.

Example 7. *For $\mathbf{p_1} = [1,0,0]^T, \mathbf{p_2} = [0,1,0]^T, \dots, \mathbf{p_6} = [0,1,1]^T, \mathbf{p_7} = [1,1,1]^T$, and $k = 3$, H can be calculated as*

$$H = \begin{pmatrix} 0 & 0 & 0 & 1 & 1\\ 0 & 0 & 1 & 0 & 1\\ 0 & 1 & 0 & 0 & 1\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 1 & 0 & 0 & 0 & 0 \end{pmatrix} \tag{17}$$

Assigning weights $\mathbf{x} = [x_1, x_2, ..., x_R]^T$ to the sets $S_1, S_2, ..., S_R$ such that $\sum_{j:\mathbf{p_i} \in S_j} x_j = v_i, \ \forall i \in \{1,2,...,m\}$ is equivalent to the existence of an $\mathbf{x}$ satisfying $H\mathbf{x} = \mathbf{v}$. Therefore, our singular conditional probability matrix is perfectly representable if and only if there exists an $\mathbf{x} \in [0,1]^R$ satisfying $H\mathbf{x} = \mathbf{v}$ where $\mathbf{v}$ is the vector of values that corresponds to our matrix. Moreover, note that the entries of $\mathbf{v}$ can not be greater than 1 and H is binary matrix, so a solution $\mathbf{x}$ for this linear system $H\mathbf{x} = \mathbf{v}$ can not have an entry greater than 1 either.

The perfect representability of our matrix is determined by the existence of $\mathbf{x} \geq 0$ satisfying $H\mathbf{x} = \mathbf{v}$. We formalize this with the following Theorem.

Theorem 2. *A matrix M is perfectly representable if and only if the corresponding vector $\mathbf{v}$ and the matrix H acquired by the above procedure satisfy*

$$\max_{\substack{\mathbf{x}\geq 0 \\ H\mathbf{x}\leq\mathbf{v}}} \mathbf{1}_R \cdot \mathbf{x} = 1 \tag{18}$$

Thus, we can use an algorithm that solves this LP to determine whether a given matrix M is perfectly representable or not after ensuring that M is a singular conditional probability matrix. If M is not a singular conditional probability matrix, then we say that it is not perfectly representable and we do not need to solve the linear program.

B. Dual LP

We can also solve the dual LP which has m variables instead:

Corollary 1. *A matrix M is perfectly representable if and only if the corresponding vector $\mathbf{v}$ and the matrix H acquired by the above procedure satisfy*

$$\min_{\substack{\mathbf{y}\geq 0 \\ H^T\mathbf{y}\geq\mathbf{1}}} \mathbf{v} \cdot \mathbf{y} = 1 \tag{19}$$

Here, our variables are $\mathbf{y} = (y_1, ..., y_m)$, thus we have $m \leq 2^k - 1$ variables.

C. Concluding Remarks

We find a necessary and sufficient condition for a conditional distribution $p_{Y|X}$ to be perfectly representable. We provide an algorithm to test for perfect representability by casting the problem as a linear program. The dual linear program has at most $m \leq 2^k - 1$ variables. Thus, while it could be exponential in the size of $\mathcal{X}$, it is linear in the size of $\mathcal{Y}$. The number of constraints depends on the Bell number R and could be exponential in k. We leave the question of whether this linear program could be solved efficiently to the future work.

Appendix

Proof of Theorem 1 (only if part):

We show that if a matrix M associated with $p_{Y|X}$ is perfectly representable, then $\mathbf{1}_k$ could be obtained from M with repeated use of PPM, VPM, and PPS operations. Since M is perfectly representable, there exists a perfect canonical functional representation W. Without loss of generality, we may assume that for each w, $p_W(w) > 0$.

First, let $Y' = (Y, W)$, so we have $Y - Y' - W$ and

$$p_{W|Y'}(w'|(y,w)) = \mathbb{1}\{w' = w\}, \tag{20}$$
$$p_{Y'|Y}((y,w)|y) = p_{W|Y}(w|y). \tag{21}$$

Note that W is a perfect functional representation for (X, Y') since $X - Y' - W$ and

$$Y' = (Y, W) = (g(X, W), W) =: g'(X, W). \tag{22}$$

Moreover, Y is a deterministic function of Y' and the matrix $p_{Y'|X}$ can be obtained from the matrix $p_{Y|X}$ by the PPS operation, so it is sufficient to show that $\mathbf{1}_k$ can be obtained from $p_{Y'|X}$ via PPM and VPM operations. As PPS is the inverse of PPM, and VPS is the inverse of VPM, it thus suffices to show that the matrix $p_{Y'|X}$ can be obtained from $\mathbf{1}_k$ via PPS and VPS operations. This is shown in the following paragraphs.

Next, fix $w \in \mathcal{W}$ and let

$$\mathcal{Y}'_w := \{(y, w) \in \mathcal{Y}' : y \in \mathcal{Y}\}. \tag{23}$$

Note that $\mathcal{Y}'$ can be partitioned into $\mathcal{Y}'_w$'s. Since W is a functional representation of (X, Y), for a fixed x and fixed w, there is a unique $(y, w) \in \mathcal{Y}'_w$ — namely $(y, w) = g'(x, w)$ — such that

$$p_{Y'|X}((y, w)|x) > 0, \tag{24}$$

and, for any $\tilde{y} \neq y$,

$$p_{Y'|X}((\tilde{y}, w)|x) = 0. \tag{25}$$

Moreover,

$$p_{Y'|X}(g'(x, w)|x) = p_W(w) > 0. \tag{26}$$

Hence, $p_{Y'|X}(g'(x, w)|x)$ is positive and does not depend on x.

Finally, for each $y' \in \mathcal{Y}'_w$, let $S_{y'}$ be the set

$$S_{y'} = \{x \in \{1, 2, ..., k\} : g'(x, w) = y'\}. \tag{27}$$

Since the set

$$\{p_{Y'|X}(y'|x) \cdot p_{W|Y'}(w|y') : y' \in \mathcal{Y}'\} \tag{28}$$

has exactly one non-zero element for any (x, w) pair, for each w, the collection $\{S_{y'} : y' \in \mathcal{Y}'_w\}$ is a partition of $\{1, 2, ..., k\}$. Now for each $y' \in \mathcal{Y}'_w$, let $\mathbf{b}_{y'}$ be a vector of size k whose j-th element is $\mathbb{1}\{j \in S_{y'}\}$. We can apply PPS operation to $\mathbf{1}_k$ with coefficients $c_w = p_W(w)$, and get vectors with patterns $\mathbf{1}_k$ and values c_w for all $w \in \mathcal{W}$. Then, for all $w \in \mathcal{W}$, we can apply VPS operation to the column with pattern $\mathbf{1}_k$ and value c_w with coefficients $\mathbf{b}_{y'}$ and get $p_{Y'|X}$ as illustrated in Figure 1. $\square$

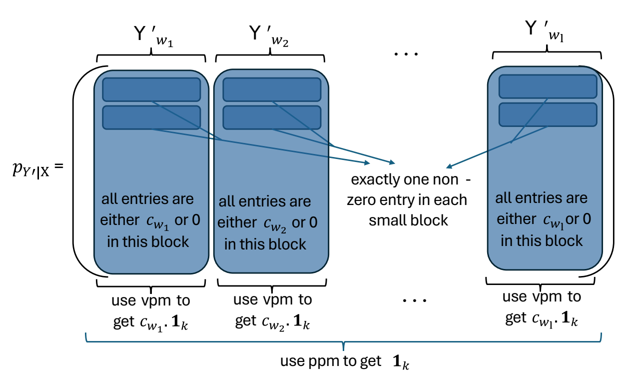

Fig. 1. Using VPM and PPM to obtain $\mathbf{1}_k$ from $p_{Y'|X}$

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